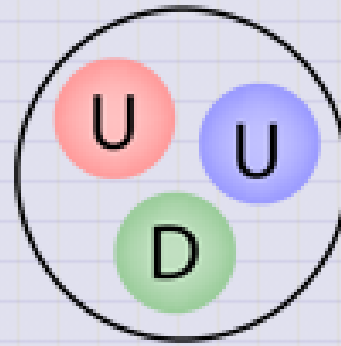
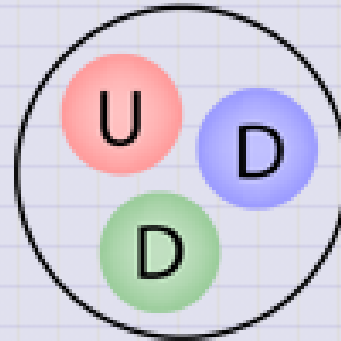


Day 16 - Conservation of Linear and Angular Momentum



Proton



Neutron

$U = \text{"up" quark} \quad +\frac{2}{3} e$
 $D = \text{"down" quark} \quad -\frac{1}{3} e$

Announcements

- Midterm 1 is due Oct 10th
- **Friday's class:** Midterm 1 Work Time ←— Take advantage of it!
- There will be no office hours on Oct 3rd (DC traveling)

Seminars this week

WEDNESDAY, October 1, 2025

Astronomy Seminar, 1:30 pm, 1400 BPS, In Person and Zoom, Host~

Speaker: Madison Brady, MSU

Title:

Zoom Link: [https://msu.zoom.us/j/93334479606?](https://msu.zoom.us/j/93334479606?pwd=OtlXPWhRPBfzYu53sl3trSJlaBYI7C.1)

[pwd=OtlXPWhRPBfzYu53sl3trSJlaBYI7C.1](https://msu.zoom.us/j/93334479606?pwd=OtlXPWhRPBfzYu53sl3trSJlaBYI7C.1)

Meeting ID: 933 3447 9606

Passcode: 825824

Seminars this week

THURSDAY, October 2, 2025

Colloquium, 3:30 pm, 1415 BPS, in person and zoom. Host ~ Witold Nazarewics

Refreshments and social half-hour in BPS 1400 starting at 3 pm

Speaker: David Dean, Thomas Jefferson National Accelerator Facility

Title: From the quarks to the cosmos

Background:

For more information and to schedule time with the speaker, see the colloquium calendar at <https://pa.msu.edu/news-events-seminars/colloquium-schedule.aspx>

Zoom Link: <https://msu.zoom.us/j/94951062663>

Password: 2002

Seminars this week

FRIDAY, October 3, 2025

Special Seminar, 10:00am, In Person Only, FRIB 1300 Auditorium

Speaker: Sudip Bhattacharyya - Tata Institute of Fundamental Research & Massachusetts Institute of Technology

Title: Thermonuclear X-ray bursts: probing neutron stars and a double-photospheric-radius-expansion

Sign up to meet with Dr. Bhattacharyya (enter your name and meeting location):

<https://docs.google.com/spreadsheets/d/1tlw765Lf8mNOeCUJp8cWfSOvQfiF4vJDOuhKKVPXH1M/edit?usp=sharing>

Seminars this week

FRIDAY, October 3, 2025

QuIC Seminar, 12:30pm, -1:30pm, 1300 BPS, Zoom only

Speaker: Balint Pato, Duke University

Title: Compass codes in quantum error correction

Full Scheule is at: <https://sites.google.com/msu.edu/quic-seminar/>

For more information, reach out to Ryan LaRose

This Week's Goals

- Understand the concept of potential energy
- Determine the equilibrium points of a system using potential energy
- Analyze the stability of equilibrium points
- Define and begin to apply conservation of linear and angular momentum

Reminders: Finding Equilibrium Points

Given a potential energy function $U(x)$, we can find the equilibrium points by setting the derivative of the potential energy function to zero:

$$\frac{dU(x^*)}{dx} = 0.$$

The stability of the equilibrium points can be determined by examining the second derivative of the potential energy function:

$$\frac{d^2U(x^*)}{dx^2} > 0? \quad \frac{d^2U(x^*)}{dx^2} < 0?$$

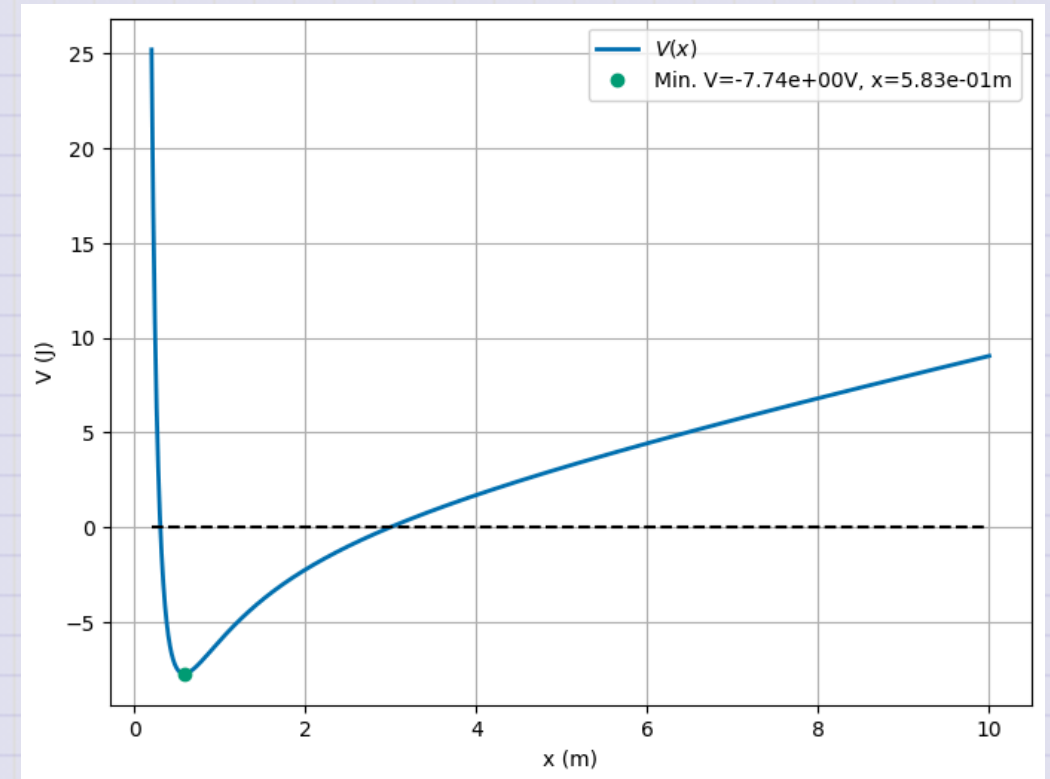
If the second derivative is positive, the equilibrium point is stable. If the second derivative is negative, the equilibrium point is unstable.

Clicker Question 16-1

Here's the graph of the potential energy function $V(x)$ that is a model of quark confinement in quantum chromodynamics.

What can you say about the equilibrium points? There is/are:

1. One stable point
2. One stable and one unstable point
3. Can't tell



Clicker Question 16-2

Here's the equation for this potential energy function:

$$V(x) = -\frac{\gamma}{x} + \frac{\delta}{x^2} + \kappa x,$$

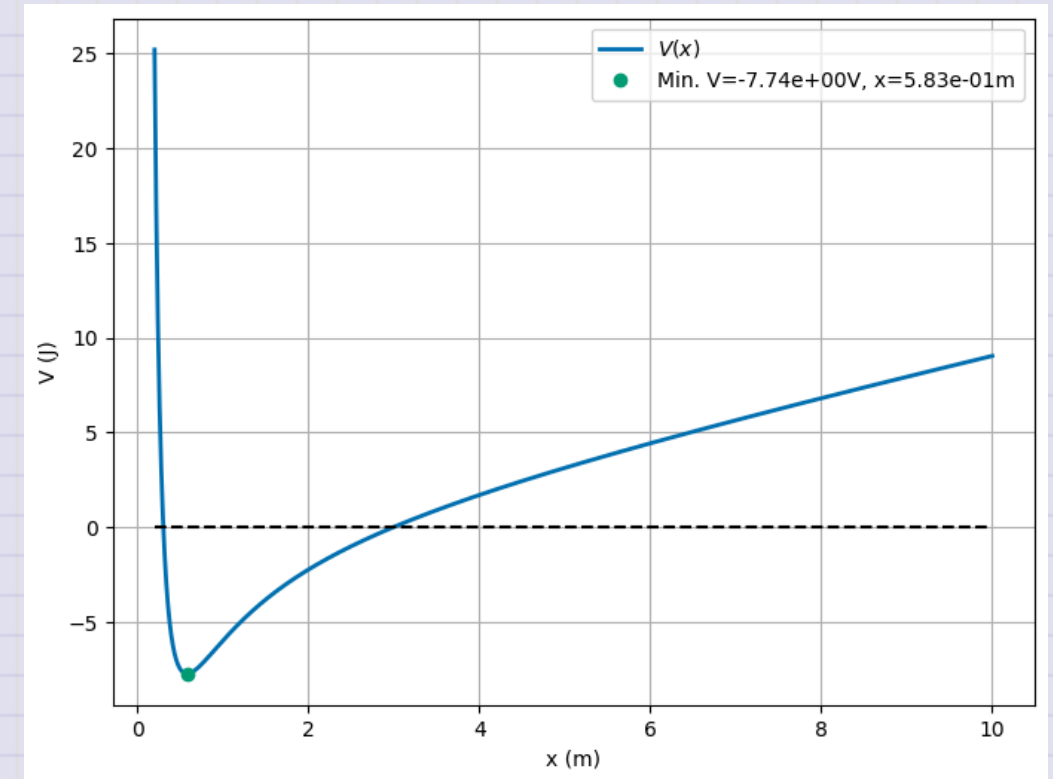
- γ , δ , and κ are constants.

What can you say about the motion of a particle with energy E ?

1. $E < 0$ 2. $E = 0$ 3. $E > 15$

Careful with #3!

Send x to ∞ : $\lim_{x \rightarrow \infty} V(x) = ?$



Clicker Question 16-3

Consider the sum of internal forces on a system of two particles:

$$\sum_i \vec{F}_i^{int} = \sum_{i=1}^{N=2} \sum_{j \neq i}^{N=2} \vec{F}_{ij} = \vec{F}_{12} + \vec{F}_{21}.$$

This sum is equal to:

1. $2\vec{F}_{12}$
2. $-2\vec{F}_{12}$
3. $\vec{F}_{12}$
4. 0
5. ???

Clicker Question 16-4

In general the sum of internal forces on a system of N particles is:

$$\sum_i \vec{F}_i^{int} = \sum_{i=1}^N \sum_{j>i}^N \left(\vec{F}_{ij} + \vec{F}_{ji} \right).$$

This sum is **always** equal to:

1. $2 \sum_i \vec{F}_i$
2. $-2 \sum_i \vec{F}_i$
3. ∞
4. 0
5. ???

Clicker Question 16-5

The change in the total angular momentum of a system of particles is given by:

$$\frac{d\vec{L}_{\text{sys}}}{dt} = \sum_i \vec{\tau}_i.$$

There is no change in the total angular momentum of a system of particles when:

1. The net external torque on the system is zero.
2. The net external force on the system is zero.
3. The net internal force on the system is zero.
4. The net internal torque on the system is zero.
5. ???

Clicker Question 16-6

We derived that the change in the total angular momentum of a system of particles is given by:

$$\frac{d\vec{L}_{\text{sys}}}{dt} = \sum_{i=1}^N \sum_{j>i}^N (\vec{r}_i - \vec{r}_j) \times \vec{F}_{ij}.$$

What geometric relationship is there between the vectors $\vec{r}_i - \vec{r}_j$ and $\vec{F}_{ij}$ if the angular momentum of the system is conserved?

1. They are parallel.
2. They are perpendicular.
3. They are anti-parallel.
4. I can't tell